Mahin
2401010123
B Tech CSE (IV)
Section A

DBMS Assign-02

Page No.
Date

Section A

Q1 Fill ups:

a) The Entity relationship model visually represents how entities relate in a database, offensive a clear blueprint of the data's organisation structure

b) In the relation model, data is organised into tables with relationships b/w entities defined through foriegn key constraints.

c) The object oriented model treats data as a collection of objects, each with unique attribute and model mirroring real-world entities.

d) Integrity const. ensure the reliability of data by enforcing ~~the~~ rules that must be adhered to maintaining the accuracy of information.

Section B

Q2 Health care system, discuss data abstraction & simplify the interactions.

Data abstraction is the process of hiding complex database details and showing only relavent info to users. In a healthcare system, data abstraction helps doctors, nurses and staff easily access patient info without dealing with complex database structures.

Page No.
Date

E.g.- Nurses can update patient vitals and treatment details.

Abstraction Levels

1) Physical level - Shows how data is stored in the database
2) Logical level - Shows the structure of the database like table and relation.
3) View level - Shows only the table data required by specified users. This makes the system easier to user and secure.

Q3. Tell benefits and challenges of data abstract°

Benefits:

→ Simplicity :- Healthcare staff can handle required data without understanding database complexity

→ Security :- Information can be restricted to authorized users only.

→ Data Integrity :- Maintain's accuracy and consistency of patient data.

→ Efficiency :- Faster access to patient records improves medical decision making.

Challenges

1) Implementation Complexity :- Designing multiple abstraction can be different.

2) Performance overhead :- Additional abstraction layers may slightly affect system performance

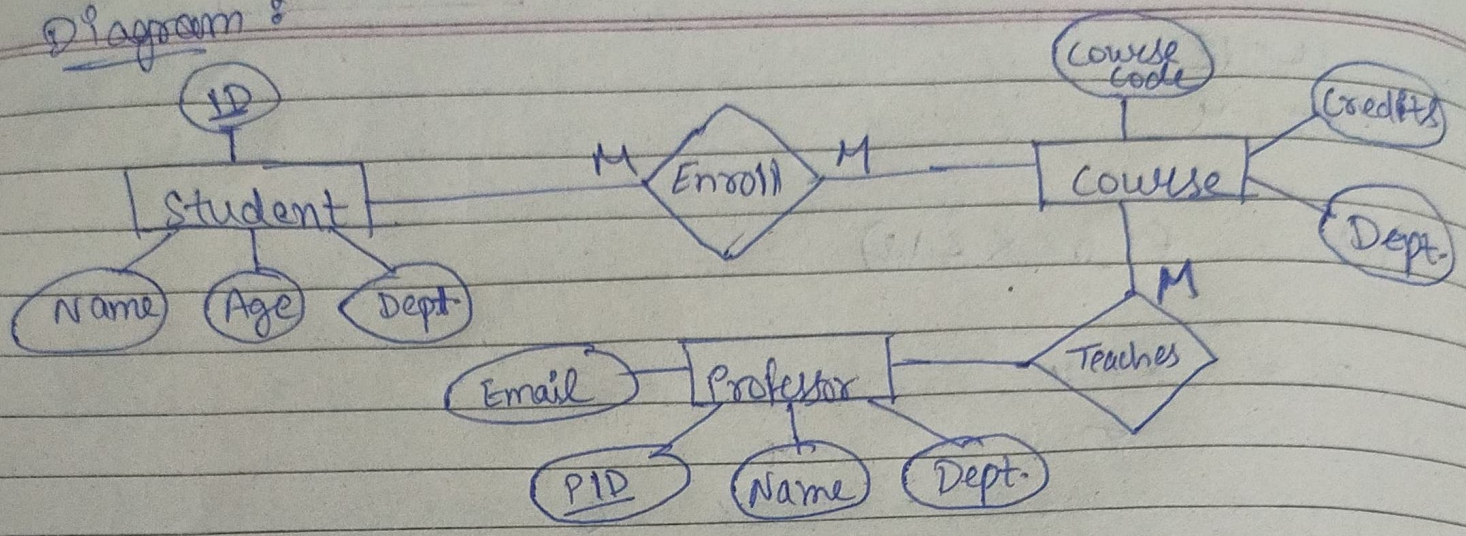
3) Difficulty in maintainance.

Section C

Q4 Design a database for university, with entities Students, course and professor, using illustration in ER. Also define their relationships

Page No.
Date

Diagram :



Entities & Attributes

(i) Student, with attributes,
- ID (PK)
- Name
- Age
- Dept.

(ii) Courses
- Course Code
- Credits
- Dept.

(iii) Professor,
- PID (PK)
- Name
- Email
- Dept.

Relationship

(i) Student - enrolls - course (M : N)
- One student can enroll in many courses
- One course can have many students

(ii) Professor - teaches - course (1 : M)
- One professor can teach multiple students
- Each course is taught by one professor.